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Skills Should Not Depend on Who Can Pay

Connectivity, affordability, disability and design as conditions of access to digital learning in Kenya and Sub-Saharan Africa

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ABSTRACT

Digital learning is often presented as a route around the scarcity of teachers, classrooms and money. The evidence supports a narrower claim. Technology widens access only when four conditions hold together: people can connect, can afford to connect, can use what they reach, and learn from it when they do. This paper reviews authoritative evidence on each condition for Kenya and Sub-Saharan Africa, drawing on international agency data, systematic reviews and randomised evaluations. It finds that the binding constraint is rarely the platform. Most people in the region who live under mobile broadband coverage still do not use it, mobile data remains several times more expensive than the international affordability target, women and people with disabilities face compounding exclusion, and devices distributed without pedagogy have repeatedly produced no learning gain. The paper derives seven design requirements for a provider that wishes access to be independent of the learner's ability to pay, sets out where Instracta's own work meets and does not yet meet them, and proposes a research agenda. The central argument is that access is a property of the whole system of delivery, and that it must be measured at the level of the excluded learner rather than the average user.

Keywords: digital learning; access to education; digital divide; affordability; inclusive design; Kenya; Sub-Saharan Africa

1. Introduction

The promise of digital learning in low- and middle-income countries rests on a simple idea: if content can be copied at near-zero cost, then skills need not be rationed by the number of teachers or the size of a family's budget. The idea is attractive to funders, governments and providers alike. It is also incomplete. A course that can be copied freely still has to reach a person, in a form that person can use, at a price that person can bear, and it has to teach something.

The scale of the problem that digital learning is asked to address is large. The 2022 update of the State of Global Learning Poverty, produced by the World Bank, UNESCO, UNICEF and partners, estimated that about 70% of ten-year-olds in low- and middle-income countries could not read and understand a simple text, up from 57% before the pandemic, and projected the figure for Sub-Saharan Africa at 89% (World Bank et al., 2022). In Kenya, the labour market consequences are visible in the youth cohort. Figures from the Kenya National Bureau of Statistics reported by the Business Daily show roughly 3.5 million people aged 15 to 34, about 19% of that age group, neither employed nor in education or training (Business Daily, 2023). Skills provision that reaches only those who can already pay does not touch this population.

This paper asks what has to be true for digital learning to reach the people it is meant for. It is a narrative review rather than a primary study. Section 2 sets out a framework of four conditions. Sections 3 to 5 review the evidence on connectivity and affordability, on gender and disability, and on whether access to technology translates into learning. Section 6 derives design requirements. Section 7 discusses limits and a research agenda. Instracta is a practitioner in this field and has an interest in the conclusions; this is stated here so that readers can weigh the argument accordingly.

2. A framework: four conditions of access

Access is often reduced to availability. The capability approach offers a more demanding standard. Sen (1999) argued that what matters for human development is not the presence of a resource but what people are actually able to do with it, a point that depends on their circumstances, abilities and the conversion of the resource into achievement. Applied to learning, a platform that exists is a resource, and a learner who completes a course and gains a usable skill has realised a capability. The distance between the two is the substance of the access problem.

We separate that distance into four conditions, each of which can fail independently.

1. Connect. The learner lives within reach of the infrastructure the service needs, whether that is a mobile network, electricity, or a physical learning point.

2. Afford. The learner, or someone acting for them, can pay the full cost of using the service, including data, devices and time, not only the course fee.
3. Use. The learner can operate the service given their language, literacy, digital skills, and any disability. This is the condition addressed by inclusive and universal design.
4. Learn. The experience leads to learning gains that are measurable and relevant. Reaching a learner with poor instruction is not access to education in any meaningful sense.

The conditions are sequential in one sense, since a learner who cannot connect never reaches the later gates, and cumulative in another, since a provider who solves three and neglects one still excludes. The framework is deliberately modest. Its purpose is to make it possible to ask, of any programme, at which gate a given learner is lost.

3. Connect and afford: the evidence on connectivity and cost

The most important pattern in the connectivity data for the region is that coverage is no longer the main gap. The GSMA's mobile connectivity series distinguishes a coverage gap, people who live outside mobile broadband coverage, from a usage gap, people who live within coverage but do not use mobile internet. In the 2024 edition, reported at the end of 2023, 27% of the Sub-Saharan population used mobile internet, 60% lived within coverage but did not use it, and 13% lived outside coverage (GSMA, 2024, as reported by Connecting Africa, 2024). Stated differently, for every person outside coverage there are more than four people under coverage who are not online. The previous edition had already reported that a quarter of the regional population used mobile internet at the end of 2022 (GSMA, 2023). Building towers is not the primary task. The primary task is making use possible and worthwhile.

Cost is one reason. The Alliance for Affordable Internet (A4AI) uses the target, adopted by the UN Broadband Commission, that 1 GB of mobile data should cost no more than 2% of average monthly income. In its 2021 analysis of 95 low- and middle-income countries, A4AI found that 1 GB in Africa cost on average 5.8% of monthly income, the highest of any region, with large variation from 0.5% in Egypt to 24.4% in the Central African Republic (A4AI, 2021). The GSMA report adds that the cost of entry-level internet-enabled handsets, relative to income, worsened significantly in 2022 after several years of improvement (GSMA, 2023). The learner's real cost of an online course is therefore a data and device cost that can exceed the course fee itself.

Kenya is more connected than the regional average, but headline numbers mislead. The Communications Authority reported 68 million active mobile subscriptions at the end of March 2024, a penetration rate of 132.1%, with 51.3 million mobile data subscriptions, of which 72.6% were on mobile broadband (reported in ITWeb Africa, 2024). Subscription counts above 100% reflect people holding more than one SIM card and should not be read as individual access.

They also say nothing about the quality of connection, the cost of data to the user, or whether the person uses the connection for learning.

Schools are in a similar position. The 2023 Global Education Monitoring Report estimated that globally only 40% of primary, 50% of lower secondary and 65% of upper secondary schools are connected to the internet, and that at least half a billion students, 31% of the world's learners and 72% of the poorest, could not be reached by remote learning during the pandemic (UNESCO, 2023). Kenya's National Digital Masterplan 2022 to 2032 sets targets to extend fibre to 40,000 public schools and learning institutions and to establish 1,450 digital innovation hubs (ITWeb Africa, 2022). The targets signal political commitment. They also show that, for the medium term, a provider cannot assume that the learner's school will be online.

4. Use: gender, disability and the problem of the average user

Averages conceal who is excluded. The GSMA's 2024 Mobile Gender Gap Report found that in Sub-Saharan Africa in 2023 women were 32% less likely than men to use mobile internet, and that 36% of women used it, leaving about 202 million women in the region offline (GSMA, 2024b, as reported by Connecting Africa, 2024b). A learning service whose usage data are not disaggregated by sex will systematically overstate who it reaches.

Disability produces a similar and often larger gap. The 2020 Global Education Monitoring Report, devoted to inclusion, found that children with disabilities make up an estimated 15% of the out-of-school population in the 14 countries with suitable data, and that children with sensory, physical or intellectual disabilities are 2.5 times more likely than their peers never to have attended school (UNESCO, 2020). On assistive technology, the WHO and UNICEF estimated in 2022 that more than 2.5 billion people need at least one assistive product and that nearly one billion are denied access, with access in some low- and middle-income countries as low as 3% of need compared with about 90% in a sample of wealthy countries (WHO and UNICEF, 2022).

The normative framework is not in doubt. Article 24 of the UN Convention on the Rights of Persons with Disabilities requires an inclusive education system at all levels (United Nations, 2006). The Constitution of Kenya guarantees every person the right to education and entitles persons with disabilities to access educational institutions and facilities integrated into society (Republic of Kenya, 2010, Arts. 43 and 54), and the Persons with Disabilities Act of 2003 provides the statutory framework (Republic of Kenya, 2003). What is less settled is the technical standard to which digital learning should be held. Two are widely used. The Web Content Accessibility Guidelines 2.2 set testable criteria for perceivable, operable, understandable and robust web content (W3C, 2023). Universal Design for Learning offers a pedagogical complement, calling for multiple means of engagement, representation, and

action and expression (CAST, 2018). A digital learning provider that does not state which standard it meets, and does not test against it, cannot claim to be accessible.

5. Learn: access to technology is not access to learning

The evidence that devices alone do not produce learning is consistent. The 2023 Global Education Monitoring Report records that Peru distributed more than one million laptops with no positive effect on learning, partly because the focus was on devices rather than on their pedagogical integration (UNESCO, 2023). Escueta, Nickow, Oreopoulos and Quan (2020), in a review of experimental research, reached a related conclusion: that technology can help, and in some cases hinder, learning, with the effect depending on how it is used.

What works is more specific. Banerjee, Cole, Duflo and Linden (2007) found in two randomised experiments in India that a computer-assisted learning programme raised mathematics scores by 0.35 standard deviations in the first year and 0.47 in the second, and that remedial teaching by community-based tutors also raised scores, whereas classmates who benefited only from smaller classes showed no gains; the gains persisted after the programme ended. Muralidharan, Singh and Ganimian (2019) evaluated a personalised, technology-aided after-school programme in urban India and found, after about 4.5 months, gains of 0.36 standard deviations in mathematics and 0.22 in Hindi for students who won a lottery for a place; they attributed the effect to instruction matched to each student's actual level, which mattered because the control group's weaker students learned at a rate close to zero. In both studies, technology served a pedagogical purpose, targeted instruction, rather than substituting for one.

Low-technology designs can also work where connectivity is constrained. Angrist, Bergman and Matsheng (2022) ran a randomised trial in Botswana of SMS messages and phone calls to parents during school closures and found that the combined treatment improved learning by 0.12 standard deviations at low cost. The result is modest in size and important in kind: a service that works over basic voice and text reaches learners that a video platform does not.

Reviews of the wider literature on learning in developing countries come to complementary conclusions. Evans and Popova (2016) found that six systematic reviews reached divergent conclusions largely because they included different studies, and that two types of programme were recommended consistently: instruction tailored to students' learning levels, delivered by teachers or adaptive software, and individualised, repeated teacher training tied to a specific task or tool. Conn (2017), in a meta-analysis of 56 articles covering 66 experiments and quasi-experiments in Sub-Saharan Africa, found that programmes changing teacher pedagogy or classroom instruction had effect sizes about 0.30 standard deviations larger than all other programme types combined. The implication for access is direct. Funding connectivity and

devices without funding the instruction that runs on them reaches the learner and then fails her at the last gate.

6. Design requirements for payer-independent access

The preceding sections imply that a provider committed to access must treat each gate as a design constraint and each barrier as a cost to be borne by the provider or its sponsors, not by the learner. We derive seven requirements. The table maps each barrier to the evidence and to the response it implies.

Table 1. Barriers to access, supporting evidence and design responses

Barrier	Evidence	Design response
Most people under coverage do not use mobile internet	60% usage gap; 13% coverage gap (GSMA, 2024)	R1. Design for low bandwidth and offline use first; treat a good connection as an enhancement
Data and devices cost more than the target	1 GB at 5.8% of monthly income in Africa against a 2% target (A4AI, 2021)	R2. Count the learner's data and device cost as part of price; use mobile-first, lightweight pages and portable packages
Licences and platforms that lock content in	UNESCO Recommendation on Open Educational Resources (2019)	R3. Prefer open licences, open-source components and exportable content so that institutions can host and adapt it without a fee
Women offline	32% less likely to use mobile internet; 36% of women online (GSMA, 2024b)	R4. Disaggregate all usage and outcome data by sex; design delivery to include women's actual channels and schedules
Disability and assistive technology gaps	2.5 times more likely never to attend school; up to 3% access to assistive products in some low- and middle-income countries (UNESCO, 2020; WHO and UNICEF, 2022)	R5. Build to WCAG 2.2 and UDL by default and publish test results
Devices without instruction produce no gain	Peru laptops; effect sizes from targeted instruction (UNESCO, 2023; Banerjee et al., 2007; Muralidharan et al., 2019)	R6. Build learning design, practice and feedback into the product, and pair it with facilitator support where possible

Barrier	Evidence	Design response
The payer is not the learner	Learning poverty of about 70% (World Bank et al., 2022) alongside data costs of 5.8% of monthly income (A4AI, 2021)	R7. Fund access through sponsors, institutions and cross-subsidy, and report how many learners participated on sponsored terms

Two of these deserve comment. R3 reflects the observation that open licensing is among the baseline indicators of the Digital Public Goods Standard, which also lists platform independence and a mechanism for extracting data (Digital Public Goods Alliance, n.d.). A learner or institution that can leave a platform with its content and data intact is less exposed to a provider's pricing or failure. R7 recognises that the problem is financial as well as technical. If the person who benefits cannot pay, someone else must, and a provider that wants access to be independent of ability to pay must say who that someone is and show that the money reached the learner.

7. Where Instracta stands, limits of the evidence, and a research agenda

Instracta's present position against these requirements is mixed, and it is better to say so. The course authoring tool exports standards-based packages that can run in other learning management systems, which supports R1 to R3, though Instracta has not yet published bandwidth benchmarks. Its Moodle theme is released as open source. The Instracta Access plugin, intended to bring a comprehensive set of accessibility features to Moodle, is planned and not yet released, so R5 is a commitment and not a record. Sponsor-funded participation, R7, is what the Instracta Reach initiative is intended to deliver, and it depends on funding that has not yet been secured. Evidence of learning gains from Instracta's own programmes is addressed in the second paper in this series.

The evidence base has limits that bear on the argument. Much of the strongest experimental evidence on technology-aided learning comes from India and from other regions, and its transfer to Kenya is plausible but untested. Kenya-specific randomised evidence is concentrated on early-grade literacy and school-based interventions rather than on adult and workforce learning, which is where most digital course providers operate. Connectivity statistics are mostly self-reported or modelled and change quickly. Finally, this paper reviews published sources; it does not report original data collection.

Four questions would most improve the evidence. First, what share of eligible Kenyan learners would complete a low-bandwidth course on a basic smartphone when data costs are met by a

sponsor, compared with when they are not? Second, what is the cost per completing learner of delivering accessible, offline-capable learning in Kenyan languages? Third, how do completion and learning gains differ by sex and disability status, and what design changes narrow the gap? Fourth, does facilitator support change learning gains enough to justify its cost? Each can be tested in pilot programmes with pre-specified outcomes and an independent evaluator.

8. Conclusion

The central claim of this paper is that access is a property of the delivery system as a whole, and that it should be measured at the margin, at the learner who is hardest to reach, rather than at the average user. The data show a region in which networks are expanding faster than use, in which data and handsets remain expensive, in which women and people with disabilities are disproportionately excluded, and in which technology produces learning only when it carries good instruction. A provider that takes these facts seriously will design for low bandwidth, price in the learner's real costs, build accessibility in from the start, and look for sponsors so that a skill does not depend on who can pay. It will also report honestly where it falls short.

Statement on method and interests

This working paper is a narrative review. Sources were identified through web and repository searches in October 2026, giving priority to international agency reports, systematic reviews, and randomised or quasi-experimental studies. Bibliographic details were checked against publisher or repository records. Some statistics were taken from secondary reports of the original publication and are marked as such in the text; readers should consult the original before relying on them. Drafting used AI assistance (Claude, Anthropic), and the author is responsible for the content. The author is the founder of Instracta eLearning Solutions, which provides the products and initiatives discussed. The paper has not been peer reviewed. Comments and corrections are welcome at the address on the Contact page.

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